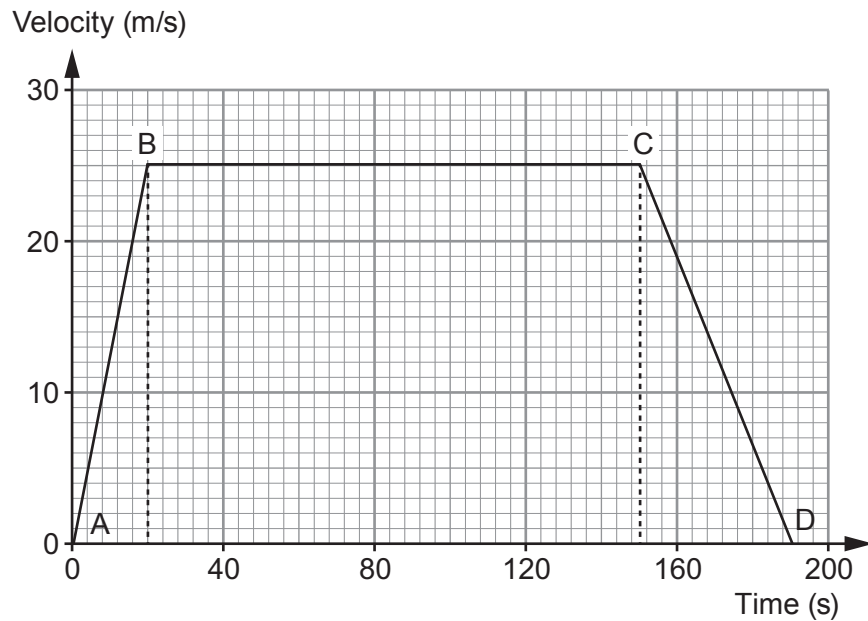


4. An empty school bus takes 190 seconds to travel between two sets of traffic lights. Its journey is shown on the velocity-time graph below.



- (a) The bus accelerates uniformly from the first set of traffic lights at A. The resultant force acting on it is 5000 N. Use the graph and equations from page 2 to calculate the mass of the empty bus. [4]

Mass = kg

- (b) The same resultant force acts on the bus which is now full of pupils. Explain how the start of the velocity-time graph (between A and B) would change. [2]

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- (c) (i) State Newton's first law of motion. [2]

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- (ii) Explain how Newton's first law applies to part **B to C** of the journey. [2]

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- (d) During its journey from A to D how long is the bus moving at a velocity greater than 12.5 m/s? [2]

Time = s

- (e) A lorry travels between the same two sets of traffic lights at a constant velocity without stopping. It takes the same time as the school bus. **On the graph opposite** draw the motion of the lorry. You may use equations from page 2 and the space below to show any calculations. [5]

Space for workings:

